

Disproof of conjecture 00000000351: D_3 is not covering-optimal in dimension 3

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1 The conjecture and the clause refuted

Statement (English, verbatim). Definition: The covering radius of the D_n lattice is determined by its coordination structure. Conjecture: For dimensions up to eight, the D_n lattices are covering-optimal, with the first non-D exception in dimension nine; and the exceptional lattice configuration is given by an E_8 extension. (D_n lattice covering optimal through dimension eight) (The Chinese text states the same.)

Reading. The conjecture is a conjunction of (A) for every dimension $n \leq 8$ the lattice D_n is covering-optimal, (B) dimension nine is the first exception, and (C) the exceptional lattice comes from an E_8 extension. We refute (A), which refutes the conjunction. The named family is $D_n = \{x \in \mathbb{Z}^n : x_1 + \cdots + x_n \text{ even}\}$ itself (not its dual D_n^*). We read “covering-optimal” as “gives the thinnest lattice covering”, i.e. minimizes the covering density Θ over all lattices of $\mathbb{R}^n$ (the lattice covering problem of Schürmann and Vallentin, Section 2), and “ D_n is” as “some similar copy $c\varphi(D_n)$ is” ($c \neq 0$ real, φ a linear isometry). The dimension used is $n = 3$, which lies in “up to eight” when the family is read as the even-sum lattices $D_n = \{x \in \mathbb{Z}^n : \sum x_i \text{ even}\}$, which are well defined and non-degenerate for every $n \geq 2$ (D_3 is the fcc lattice). Under the root-system classification convention that reserves type D_n for $n \geq 4$, dimension 3 would be excluded; this package does not supply a counterexample in the range $4 \leq n \leq 8$. We refute:

- (R1) some similar copy of D_3 has covering density at most that of every lattice in $\mathbb{R}^3$ (`not_DCoveringOptimal_three`); hence the claim fails for every range of dimensions containing 3, in particular $3 \leq n \leq 8$ (`conjecture_351_false`);
- (R2) (stronger) one explicit lattice, the body-centred cubic lattice, has *strictly* smaller covering density than *every* similar copy of D_3 (`coveringDensity_bcc_lt_similar_D3`); in particular $\Theta(\text{bcc}) < \Theta(D_3)$ (`coveringDensity_bcc_lt_D3`).

Not addressed. Clauses (B) and (C) are not used. The Definition line (“determined by its coordination structure”) is descriptive and imposes nothing formal; we use the standard covering radius. The reading “optimal among all (also non-lattice) sphere coverings” implies the lattice reading (the bcc lattice covering is itself a covering), but the density of a general covering is not formalized. The packing-covering ratio μ/λ and the dual family D_n^* are not treated.

2 Definitions (as in the Lean file)

We work in $\mathbb{R}^n = \text{EuclideanSpace } \mathbb{R} \text{ (Fin } n)$ with Lebesgue measure `vol` (Mathlib’s `volume`) and Euclidean distance.

Definition 1 (lattice; `Lattice n`). A $\mathbb{Z}$ -submodule $\Lambda \subset \mathbb{R}^n$ that is discrete and spans $\mathbb{R}^n$ over $\mathbb{R}$ (Mathlib’s `DiscreteTopology` and `IsZLattice R`). Its covolume `covol` Λ is Mathlib’s `ZLattice.covolume`, the volume of a fundamental domain. Wikipedia (Lattice (group)): “The n -dimensional volume of this fundamental domain is sometimes called the covolume of the lattice: it is invariant for any basis, and may be computed as $d(\Lambda) = |\det(M)|$ ”.

Definition 2 (covering radius; `coveringRadius`). For $S \subseteq \mathbb{R}^n$, $\mu(S) = \inf\{r \geq 0 : \forall x \in \mathbb{R}^n \exists l \in S, \|x - l\| \leq r\}$. Wikipedia (Delone set): “The covering radius, R , of X is the smallest distance such that every point of M is within distance R of at least one point in X ; that is, R is the smallest radius such that closed balls of that radius centered at the points of X have all of M as their union.” Schürmann–Vallentin: “the covering radius $\mu(L)$ by $\mu(L) = \min\{\mu : L + \mu B^d \text{ is a lattice covering}\}$ ”. Convention: in Lean $\inf \emptyset = 0$; every use below is for a set for which a covering radius $R < \infty$ is proved first, so the value is the genuine infimum.

Definition 3 (covering density; `coveringDensity`). $\Theta(\Lambda) = \text{vol}(\overline{B}(0, \mu(\Lambda))) / \text{covol } \Lambda$. Schürmann–Vallentin: “the covering density $\Theta(L) = \frac{\mu(L)^d}{\det(L)} \cdot \kappa_d$, $\kappa_d = \text{vol } B^d$, is invariant with respect to scaling of L ” and “Problem 2.1. (Lattice covering problem) For a given d , determine $\Theta_d = \min_L \Theta(L)$, where L runs over all d -dimensional lattices.” The two formulas agree: Mathlib’s `EuclideanSpace.volume_closedBall_fin_three` gives $\text{vol } \overline{B}(0, r) = r^3 \cdot 4\pi/3$ (lemma `vol_closedBall`).

Definition 4 (D_n , similar copies, optimality). `Dlat n` = $\{x \in \mathbb{Z}^n : \sum x_i \text{ even}\}$. Wikipedia (Root system): “The D_n root lattice – that is, the lattice generated by the D_n roots – consists of all integer vectors whose components sum to an even integer.” `IsSimilarCopy M Q`: $Q = \{c\varphi(x) : x \in M\}$ as sets, for some real $c \neq 0$ and linear isometry φ . `DCoveringOptimal n`: there is a $\mathbb{Z}$ -submodule Q that is a similar copy of D_n with $\Theta(Q) \leq \Theta(L)$ for every lattice L of $\mathbb{R}^n$.

The body-centred cubic lattice `bcc` is the $\mathbb{Z}$ -span of $e_1, e_2, h = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$; Lean proves `bcc = \{z + \frac{e}{2}(1, 1, 1) : z \in \mathbb{Z}^3, e \in \mathbb{Z}\} = \mathbb{Z}^3 \cup (\mathbb{Z}^3 + h)` (`mem_bcc_iff`).

3 Proof

Lemma 5 (`D3_covers`, `D3_deep_hole`, `similar_D3_bounds`). (a) Every $y \in \mathbb{R}^3$ is within distance 2 of D_3 . (b) $\|(1, 0, 0) - m\| \geq 1$ for all $m \in D_3$. (c) If $Q = c\varphi(D_3)$ then $\mu(Q) \geq |c|$ and $\text{covol } Q = 2|c|^3$.

Proof. (a) Take $m_i = 2 \text{round}(y_i/2)$; then $\sum m_i$ is even and $|y_i - m_i| \leq 1$, so $\|y - m\|^2 \leq 3 \leq 4$. (b) Write $m = z \in \mathbb{Z}^3$ with $z_1 + z_2 + z_3$ even. If $z_2 \neq 0$ or $z_3 \neq 0$ then $\|(1, 0, 0) - z\|^2 \geq 1$; otherwise z_1 is even, so $z_1 \neq 1$ and $(1 - z_1)^2 \geq 1$. (c) Distances scale: $\|c\varphi a - c\varphi b\| = |c| \|a - b\|$. By (a), balls of radius $2|c|$ around Q cover $\mathbb{R}^3$ (apply (a) to $\varphi^{-1}(c^{-1}x)$), so $\mu(Q)$ is a genuine infimum; by (b) the point $c\varphi(1, 0, 0)$ has distance $\geq |c|$ from every point of Q , so every admissible radius is $\geq |c|$. For the covolume, D_3 is the $\mathbb{Z}$ -span of the rows of $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$ (`Dlat_eq_span`), of determinant 2; Q is the $\mathbb{Z}$ -span of their images, of determinant $c^3 \det \varphi \cdot 2$ with $|\det \varphi| = 1$ (`abs_det_isometry`), and $\text{covol}(\mathbb{Z}\text{-span of a basis } b) = |\det b|$ (`covolume_span`, from `ZLattice.covolume_eq_det_mul_measureReal`). $\square$

Lemma 6 (`round_pair`, `bcc_covers`, `covolume_bcc`). Every $x \in \mathbb{R}^3$ is within distance $5/8$ of `bcc`, and $\text{covol}(\text{bcc}) = \frac{1}{2}$.

Proof. For $y \in \mathbb{R}$ let $u = y - \lfloor y \rfloor \in [0, 1)$. If $u \leq \frac{1}{2}$ take $a = b = \lfloor y \rfloor$: then $(y - a)^2 + (y - b - \frac{1}{2})^2 = 2u^2 - u + \frac{1}{4} \leq \frac{1}{4}$. Otherwise take $a = \lfloor y \rfloor + 1$, $b = \lfloor y \rfloor$: the sum is $2u^2 - 3u + \frac{5}{4} \leq \frac{1}{4}$ since $(u - \frac{1}{2})(1 - u) \geq 0$. Apply this to each coordinate of x to get $A = (a_i) \in \mathbb{Z}^3$ and $B = (b_i + \frac{1}{2}) \in \mathbb{Z}^3 + h$, both in bcc (mem_bcc), with $\|x - A\|^2 + \|x - B\|^2 \leq \frac{3}{4}$. Hence one of them is at squared distance $\leq \frac{3}{8} \leq \frac{25}{64}$. The basis matrix with rows e_1, e_2, h has determinant $\frac{1}{2}$. $\square$

Theorem 7 (`coveringDensity_bcc_le`, `coveringDensity_similar_D3_ge`, `coveringDensity_bcc_lt_similar_D3`). $\Theta(\text{bcc}) \leq \frac{125\pi}{192} < \frac{128\pi}{192} = \frac{2\pi}{3} \leq \Theta(Q)$ for every similar copy Q of D_3 .

Proof. By Lemma 6, $0 \leq \mu(\text{bcc}) \leq \frac{5}{8}$, so $\Theta(\text{bcc}) \leq \frac{(5/8)^3 \cdot 4\pi/3}{1/2} = \frac{125\pi}{192}$. By Lemma 5(c), $\Theta(Q) = \frac{\mu(Q)^3 \cdot 4\pi/3}{2|c|^3} \geq \frac{|c|^3 \cdot 4\pi/3}{2|c|^3} = \frac{2\pi}{3}$. $\square$

Taking $c = 1$, $\varphi = \text{id}$ gives $\Theta(\text{bcc}) < \Theta(D_3)$. If some similar copy Q of D_3 satisfied $\Theta(Q) \leq \Theta(L)$ for every lattice L , then $L = \text{bcc}$ would contradict the theorem. So D_3 is not covering-optimal, and (A) fails. This is consistent with the literature: Schürmann–Vallentin write “The optimality of the body centered cubic lattice [...] was first proven by Bambach [Bam54b]”. That result is not used; the argument above is self-contained.

4 Lean formalization

File `lean/Conjecture351/Basic.lean` (354 lines, `import Mathlib` only, namespace `C351`). Main statements:

```
theorem coveringDensity_bcc_lt_similar_D3 (Q : Submodule Z (Rn 3))
  (hQ : IsSimilarCopy (Dlat 3) Q) :
  coveringDensity bcc.L < coveringDensity Q
theorem coveringDensity_bcc_lt_D3 :
  coveringDensity bcc.L < coveringDensity (Dlat 3)
theorem not_DCoveringOptimal_three : not DCoveringOptimal 3
theorem conjecture_351_false :
  not (forall n in Finset.Icc 3 8, DCoveringOptimal n)
```

(ASCII transcription; `bcc.L` is the field `bcc.Λ`.) `#print` axioms for each of these, and for `mem_bcc_iff`, reports only `propext`, `Classical.choice`, `Quot.sound`. There is no `sorry`, no `native_decide` and no added axiom.

Credit. `Dlat`, `IsSimilarCopy`, `covolume_span`, `basisOfDet` and `abs_det_isometry` are adapted (dimension 7 to 3) from the accepted package for conjecture 00000000238 (GPL-3.0).

Sources

- A. Schürmann, F. Vallentin, Computational approaches to lattice packing and covering problems, arXiv:math/0403272, Section 2 and Section 3. <https://arxiv.org/abs/math/0403272>
- Wikipedia, Delone set. https://en.wikipedia.org/wiki/Delone_set
- Wikipedia, Root system. https://en.wikipedia.org/wiki/Root_system
- Wikipedia, Lattice (group). [https://en.wikipedia.org/wiki/Lattice_\(group\)](https://en.wikipedia.org/wiki/Lattice_(group))